

ML based Smart Wearable Defibrillator with the integration of SoC

Abstract—The evolution of wearable medical devices marks a significant advancement in personalized healthcare and real-time monitoring, offering substantial benefits in patient care and emergency medical scenarios. This article presents the design and implementation of a Machine Learning-based Smart Wearable Defibrillator with System-on-Chip (SoC) integration. The architecture is systematically divided into modular sub-components, allowing for individual design and testing to ensure precise functionality before integration into the complete system. The primary objective is to leverage Machine learning algorithms to automatically administer therapeutic shocks when the heartbeat rate deviates from predetermined threshold values, thus eliminating the need for manual intervention or hospital assistance and potentially reducing mortality rates. Key components include a low-pass filter to suppress frequencies above 1.67 Hz, corresponding to standard heartbeat frequencies, and an automatic gain control amplifier to enhance signal levels. The processed signal is then analyzed by a comparator to produce the necessary output. This comprehensive approach, supported by SoC technology and Machine learning, aims to improve the reliability and effectiveness of wearable defibrillator technology, thereby enhancing patient outcomes in emergency situations.

Index Terms—Wearable Cardioverter Defibrillator (WCD), System on Chip (SoC), Machine Learning, Heartbeat Monitoring, Signal Processing

I. INTRODUCTION

Cardiovascular diseases (CVDs) remain the leading cause of death worldwide, responsible for over 17 million deaths annually, according to the World Health Organization [1]. The burden of these diseases highlights the urgent need for continuous innovation in diagnostic and therapeutic technologies. Among these innovations, the development of smart wearable devices equipped with machine learning algorithms has shown significant promise in enhancing real-time monitoring and emergency interventions for cardiac events [2]. This paper introduces an innovative concept of a Machine learning-based smart wearable defibrillator that integrates System-on-Chip (SoC) technology, aiming to provide enhanced cardiac care through advanced computational capabilities and efficient power management. Wearable defibrillators have revolutionized cardiac care by offering portable, non-invasive solutions for detecting and treating arrhythmias [3]. These devices continuously monitor the heart's electrical activity and can automatically deliver life-saving defibrillation shocks when necessary [4]. The integration of machine learning algorithms into these devices allows for improved accuracy in arrhythmia detection and personalized treatment plans based on real-time data analysis [5]. Recent studies have demonstrated the effectiveness of machine learning models in predicting and diagnosing cardiac events, showing superior performance

compared to traditional methods [6] [7]. The incorporation of SoC technology into wearable medical devices offers several advantages, including miniaturization, power efficiency, and advanced computational capabilities [8]. SoCs integrate all the necessary components of a computer or electronic system onto a single chip, making them ideal for use in compact and lightweight wearable devices [9]. This integration is crucial for continuous health monitoring and timely therapeutic responses, as it enables the device to operate efficiently without the need for frequent recharging [10]. Microwave therapy, which involves the use of microwave radiation to generate therapeutic effects such as tissue heating, has shown potential in various medical applications, including oncology and pain management [11]. Integrating this therapy into a wearable defibrillator could offer innovative treatment modalities, enhancing the therapeutic capabilities of the device [12]. The use of non-Foster elements in RF and microwave devices can further improve design flexibility and device performance, which is pertinent to developing advanced therapeutic devices [13]. Self-powered wearable electronic systems enabled by microwave-processed thermoelectric materials can harness body heat to generate electricity, providing a sustainable energy solution for continuous health monitoring [14]. Additionally, near-field coupled wireless microwave sensors can enhance the functionality of smart wearable defibrillators by providing cost-effective and reliable detection of dielectric materials [15]. In this paper, we proposed the development of a machine learning-based smart wearable defibrillator with SoC integration represents a significant advancement in cardiac care technology. By leveraging the latest in semiconductor technology, microwave therapy, and machine learning algorithms, this device aims to provide continuous monitoring and effective treatment for patients at risk of sudden cardiac arrest. The combination of these cutting-edge technologies promises to improve patient outcomes and save lives.

II. LITERATURE REVIEW

The integration of machine learning algorithms into wearable medical devices represents a significant advancement in the field of cardiac care. These smart wearable defibrillators equipped with System-on-Chip (SoC) technology are poised to revolutionize how we monitor and respond to cardiac events. The prevalence of cardiovascular diseases (CVDs) as the leading cause of mortality worldwide, accounting for over 17 million deaths annually, underscores the need for innovative solutions [1]. The ability of machine learning models to accurately predict and diagnose cardiac events has

been well-documented, showcasing their superior performance compared to traditional methods [2] [5]. Wearable defibrillators have proven to be effective in providing continuous cardiac monitoring and delivering life-saving interventions when necessary [?]. The integration of machine learning into these devices enhances their capability to detect arrhythmias with high precision, allowing for timely and personalized therapeutic responses [6]. Recent research has demonstrated that convolutional neural networks (CNNs) and recurrent neural networks (RNNs) can achieve cardiologist-level accuracy in detecting arrhythmias, highlighting the potential of these technologies in clinical settings [7]. SoC technology plays a critical role in the development of these advanced wearable devices. By integrating all necessary components onto a single chip, SoCs enable the miniaturization and power efficiency required for continuous monitoring and real-time data processing [8]. This technology is crucial for ensuring that wearable defibrillators are compact, lightweight, and capable of long-term operation without frequent recharging [9]. The use of microwave-processed thermoelectric materials in these devices can further enhance their performance by enabling efficient self-powering through body heat [10]. Microwave therapy, which involves the application of microwave radiation to generate therapeutic effects such as tissue heating, has shown promise in various medical applications. Its integration into wearable defibrillators could provide innovative treatment modalities, improving the therapeutic outcomes for patients at risk of sudden cardiac arrest [11]. Research into the use of non-Foster elements in RF and microwave devices has shown that these components can significantly improve the design flexibility and performance of therapeutic devices [13]. Self-powered wearable electronic systems are another important area of development. These systems leverage advanced thermoelectric materials to harness body heat, providing a sustainable energy source for continuous health monitoring [14]. The incorporation of near-field coupled wireless microwave sensors into smart wearable defibrillators can enhance their detection capabilities, making them more effective in real-time monitoring [15]. Overall, the combination of machine learning, SoC technology, and microwave therapy in smart wearable defibrillators represents a significant advancement in cardiac care technology. This integrated approach leverages the latest innovations in semiconductor technology and machine learning to provide enhanced monitoring and treatment options for patients, potentially improving outcomes and saving lives.

III. PROPOSED DESIGN MODEL

A Smart Wearable Defibrillator (SWD) is an advanced medical device designed to monitor heart activity and deliver therapeutic shocks in case of life-threatening arrhythmias, such as ventricular fibrillation or tachycardia. The integration of smart technology with wearable devices has significantly enhanced the capabilities and usability of defibrillators, making them more accessible and efficient for patients at risk of sudden cardiac arrest. One of the critical components of SWDs is the System On-Chip (SoC) design for signal

processing units, which allows for efficient monitoring and response to cardiac events. This design is crucial in addressing cardiac problems exacerbated by modern lifestyle factors, offering a reliable and responsive solution for patients [16]. The integration of artificial intelligence (AI) and the Internet of Things (IoT) in SWDs has significantly advanced remote health monitoring capabilities. These technologies enable the continuous monitoring of vital signs such as pulse rate, body temperature, blood pressure, and blood oxygen saturation. The data collected through wearable sensors can help detect diseases like diabetes, heart disease, and gait disturbances [17].

A. Smart Wearable Defibrillator

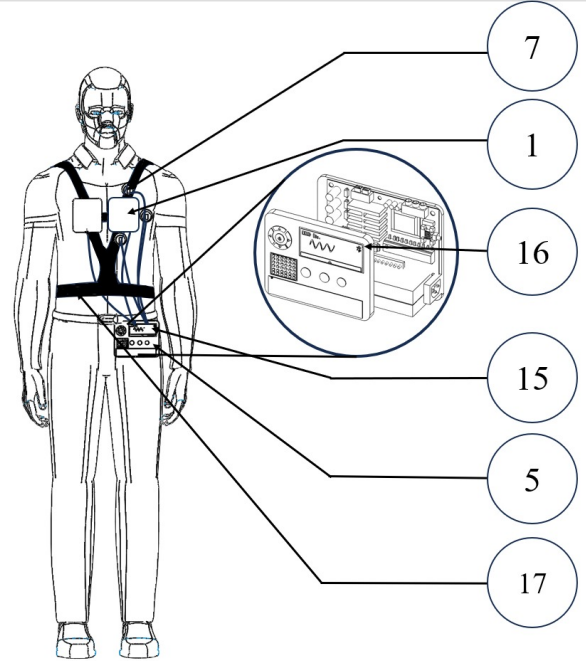


Fig. 1. Smart Wearable Defibrillator

1) *Electrode adhesive patches*: Module No.7 in Fig. 1 denotes Electrode adhesive patches. They securely pass Ag/AgCl electrodes to the patient's skin, ensuring consistent ECG signal acquisition. These patches maintain stable electrode-body contact, reducing motion artifacts and interference, thus improving signal quality. Designed with wearer comfort in mind, they use hypoallergenic materials and non-irritating adhesives for extended wearability. Whether disposable or reusable, these patches enhance the durability and reliability of the SoC device, facilitating continuous monitoring and effective arrhythmia management.

2) *Electrodes*: Module No.1 in Fig. 1 is a wearable dry electrodes for bio-sensing applications, eliminating the need for gel and ensuring comfortable ECG signal measurement. It has a conductive polymer layer between electrodes to reduce skin-electrode impedance. Carbon nanotubes provide conductivity. Unlike wet electrodes, dry electrodes avoid gel drying issues, ensuring accurate ECG readings. Once collected, ECG signals undergo further processing.

3) *Bluetooth module*: Module No.16 in Fig. 1 illustrates bluetooth module in the System On-Chip (SoC) device for the smart wearable defibrillator enables wireless communication with external devices. Using Bluetooth Low Energy (BLE) technology, it transmits real-time ECG signals and device status updates to smartphones, tablets, or medical systems. This allows remote monitoring by healthcare professionals, enhancing patient care and intervention efficiency. The module supports bidirectional communication, enabling the defibrillator to receive commands or updates, ensuring adaptability within the healthcare ecosystem.

4) *Display*: Module No.15 in Fig. 1, the display is essential for providing real-time feedback and crucial information about device status and the patient's condition. It shows operational status, battery level, and ECG waveform, enabling continuous monitoring and prompt identification of abnormalities. Visual alarms alert users to critical situations like shockable rhythms or low battery levels, ensuring timely response. The display also allows menu navigation, settings adjustment, and provides instructional messages for effective operation. Additionally, it supports data analytics for reviewing historical ECG data, aiding healthcare providers in assessing patient trends and treatment effectiveness.

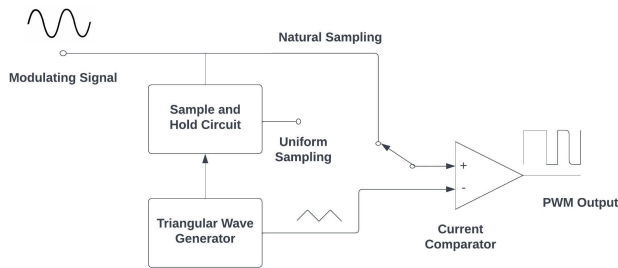


Fig. 2. PWM Circuit Block

5) *Defibrillating Unit*: Module No.5 in Fig. 1 shows a defibrillating unit that delivers an electric shock to the heart in case of a cardiac arrest. It contains a biphasic voltage waveform generator. Since biphasic waveforms were found to have much higher efficiency as per the discussion in the previous sections, the biphasic waveform is generated using Pulse Width Modulation (PWM). The PWM circuit was designed to have a low frequency biphasic waveform. The human ECG signal consists of PQRST waves. The frequency range of P and T waves are 0.5 and 10 Hz respectively and QRS range between 4 and 20Hz. The standard frequency range for ECG signals ranges between 0.05 to 100 Hz. A bandpass filter allows signals only between two specific frequencies. But a low pass allows all signals with frequencies lower than cutoff frequency attenuates higher frequencies. The filter used here is a low pass filter. The input of the filter is the ECG signal. It attenuates the noise and produces the desired response. AC analysis of the filter is obtained with the cutoff frequency of 100Hz. The ECG signal will be in a few millivolts which

is not appreciable for the generating shocks. Hence there is a need to amplify the input signal. Automatic gain amplifiers are employed as the input ECG signals vary over a wide dynamic range. The second waveform depicts the input filtered ECG signal with amplitude of 3mV. The first waveform is the output signal with amplitude of 15V. The peak-to-peak 6mV input signal is amplified to 30V peak-to-peak with frequency of 100Hz. It produces constant gain irrespective of the variations in the input signal. The input signal and the reference signal are set to 7.2V and 3V respectively. Here the input signal is an AC sine signal and the reference signal is the DC signal. The input signal is the ECG signal of the person that is being monitored by the device and the reference signal is the standard ECG signal of a normal human being. The comparator now compares the input and the reference signal. The comparator is a circuit which compares the input signal with the reference signal and produces the binary signal as the output. During the positive half cycle, when the input signal is higher than the reference signal the output will be a binary low. During the negative half cycle, when input is lesser than the reference signal the output will be a binary high with an amplitude of 6.6V. When the output of the comparator is low, a shock will be generated which will help the heart to regain its natural pace and overcome the problem of cardiac arrest.

6) *Adjustable belts*: Module No.17 in Fig. 1 represents adjustable belts integrated into the System On-Chip (SoC) device for the smart wearable defibrillator enhance usability and comfort. They provide a secure, customized fit, ensuring the device stays in place without discomfort or restriction during daily activities. With flexibility in sizing and fit, the belts accommodate various body types, optimizing user experience and promoting long-term compliance.

B. System on Chip (SoC)

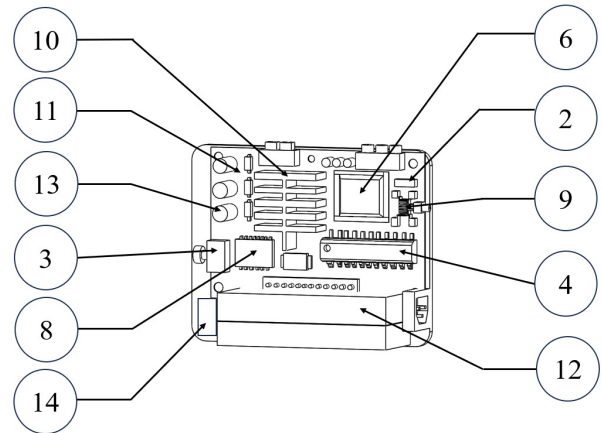


Fig. 3. Integrated System on Chip

1) *Heat Sink*: Module No.10 in Fig. 3 is a heat sink for managing thermal performance. As the SoC operates, it generates heat from current flow and transistor switching,

potentially affecting device reliability. The heat sink, typically made of materials like aluminum or copper, dissipates this heat into the environment using its large surface area. It's directly attached to the ICs or the SoC package using thermal interface materials like thermal paste or pads to ensure efficient heat transfer.

2) *Microcontroller*: The microcontroller (Module No.11 in Fig. 3) serves as the brain, handling essential functions like real-time monitoring, signal processing, and classification of ECG rhythms. It processes raw ECG signals by filtering, amplifying, and digitizing them for accurate data acquisition. Moreover, it implements machine learning frameworks to differentiate shockable and non-shockable conditions accurately. The microcontroller also manages component operations, optimizes power consumption, and interfaces with user inputs and displays for seamless interaction.

3) *Capacitors*: Capacitors (Module No.13 in Fig. 3) accumulate charge from a power source like a rechargeable battery during the charging phase. When a shock is required, capacitors discharge rapidly, releasing stored energy as a controlled electric pulse through defibrillation electrodes to the patient's heart. Capacitance value and voltage rating are carefully chosen to meet energy requirements and safety standards, ensuring effective shock delivery while minimizing risks. Charge control circuitry regulates charging and discharging, ensuring precise energy delivery and optimizing defibrillator performance.

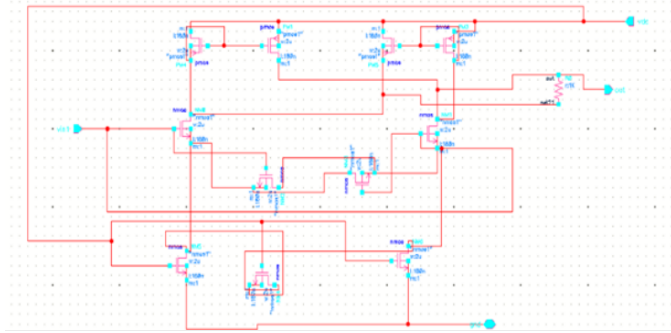


Fig. 4. Automatic gain control amplifier circuit

4) *Automatic Gain Control Amplifier*: An automatic gain amplifier (Module No.3 in Fig. 3) maintains a consistent gain regardless of the input's amplitude. Since the ECG signal has a maximum peak of a few millivolts, it may not be sufficient for the comparator block. To address this issue, a differential pair amplifier, combined with a modified Gilbert cell, introduces a particular gain to a dynamic range of inputs. The Gilbert cell serves as a second stage of amplification. To prevent any potential body effect, the PMOS body is linked to the source while the NMOS body is attached to a low potential.

5) *Signal conditioning circuitry*: This circuitry (Module No.8 in Fig. 3) consists of amplifiers, filters, and analog-to-digital converters (ADCs) designed to enhance the quality and accuracy of the ECG data before further processing. Amplifiers ensure optimal signal strength, filters eliminate unwanted

noise and artifacts, and ADCs digitize the analog signals for subsequent digital processing by the SoC unit. Together, these components enable reliable and accurate analysis of the ECG signals, enhancing the performance and effectiveness of the wearable defibrillator in detecting and responding to cardiac abnormalities.

6) *Charge control circuitry*: Charge control circuitry (Module No.14 in Fig. 3) manages the crucial process of charging and discharging capacitors for delivering electric shocks to the heart. It includes a charging circuit to regulate input voltage and current, preventing overcharging, and a discharging circuit for controlled energy release. Safety features such as voltage monitoring and fault detection ensure optimal performance and safety. The circuitry interfaces with the SoC's microcontroller for commands and user interfaces, providing real-time feedback on charging status and safety alerts.

7) *Machine learning (ML) Unit*: Deep neural networks, an extension of neural networks, are pivotal in computer-aided diagnosis for various diseases. They automatically extract features and classify data, outperforming classical machine learning methods. In ECG signal analysis, recurrent and convolutional neural networks are commonly used. Machine learning networks enhance system robustness compared to manual feature extraction methods. However, they require a large training dataset for optimal performance. Despite this limitation, machine learning algorithms are more effective in detecting shockable ECG signals. This unit will be briefly explained in section IV.

8) *Low Pass Filter Circuit*: A low pass filter (Module No.2 in Fig. 3) permits all frequencies to pass through until the cutoff frequency and suppresses other frequencies as shown in Fig. 5. The standard frequency range for an ECG signal is typically between 0.05 and 100Hz. Therefore, to simplify the band pass filter design and eliminate unnecessary noise and frequencies beyond 100Hz, a low pass filter is preferred.

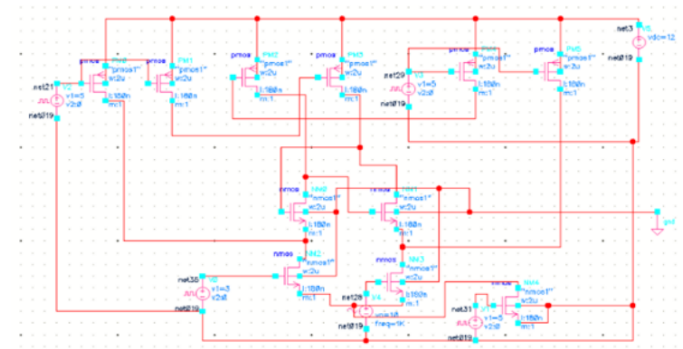


Fig. 5. Low pass filter circuit

9) *Microwave therapy module*: Microwave therapy (Module No.9 in Fig. 3) harnesses high-frequency electromagnetic waves generated by a microwave generator to selectively heat and destroy specific tissues within the body. This innovative therapeutic approach offers a minimally invasive alternative for treating conditions like cancer and chronic pain. The

microwave generator produces electromagnetic waves within the microwave frequency range, typically between 915 MHz to 2.45 GHz. Microwave therapy integrated into a System On-Chip (SoC) device for a smart wearable defibrillator can enhance treatment of cardiac arrhythmias by providing targeted tissue ablation, reducing the need for high-energy shocks, and offering a non-invasive, more comfortable alternative to traditional defibrillation. This integration allows for real-time monitoring and adjustment, improving therapy effectiveness and patient compliance. Additionally, the device can continuously manage heart rhythms, collecting detailed data to refine treatments and personalize therapy, all while ensuring efficient power management and user-friendly operation. This advancement in wearable defibrillator technology can significantly improve patient outcomes and quality of life.

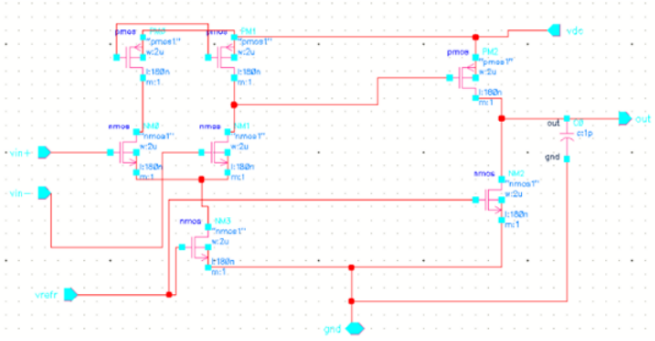


Fig. 6. Comparator circuit

10) Comparator: As the name suggests, a comparator (Module No.4 in Fig. 3) circuit generally compares two voltage levels and provides a high level '1' or a low level '0' at the output in accordance with the comparison result. The open loop comparator is similar to an operational amplifier working in saturation as illustrated in Fig. 6. Thus when the peak of the ECG signal exceeds the threshold voltage level a low level is obtained at the output and a high level appears when the peak of the ECG signal doesn't exceed the threshold level. The advantage of the circuit is that it provides high resolution. To avoid body affect the body of the PMOS is connected to the source and the body of the NMOS is connected to low potential.

11) Battery pack: In the System on Chip (SoC) device for a smart wearable defibrillator, while the battery pack is typically not integrated into the SoC itself, the SoC manages power consumption and regulates power from an external source. It Module No.12 in Fig. 3 ensures stable operation, implements low-power modes to conserve energy, and monitors battery levels to prevent depletion. The SoC may also include charging control for rechargeable batteries, optimize power efficiency, and provide emergency power reserve for critical operations like defibrillation, ensuring efficient and reliable device operation with extended battery life.

IV. MACHINE LEARNING MODEL

A. Preprocessing

To achieve high classification accuracy, robust preprocessing algorithms are crucial. This initial step involves reducing noise in the ECG signal by smoothing it, suppressing drift and baseline wander, and preparing it for subsequent processes. The following methods are commonly employed to reduce signal noise: (i) Second-order low-pass and high-pass Butterworth filtering, (ii) Daubechies wavelet 6 (db6), and (iii) Orthogonal wavelet filter. Previous studies have also segmented ECG signals into beats or segments of different durations (e.g., 2 and 5 s) before extracting features. The Pan-Tompkins algorithm, which detects ECG R-peaks for segmentation, is a popular algorithm used for this purpose. For this preprocessing, we classified the ECG signals as shown in Table I

TABLE I
DESCRIPTION OF FEATURES IN THE DATASET

Feature	Description
Age	Age of the patient
Sex	Sex of the patient
CP	Chest pain type: 0 = typical angina, 1 = atypical angina, 2 = non-anginal pain, 3 = asymptomatic
Trestbps	Resting blood pressure in mm Hg
Chol	Cholesterol in mg/dl
Exng	Exercise induced angina: 1 = yes, 0 = no
Fbs	Fasting blood sugar $\geq$ 120 mg/dl: 1 = true, 0 = false
Restecg	Resting electrocardiographic results: 0 = normal, 1 = having ST-T wave abnormality, 2 = showing probable or definite left ventricular hypertrophy
Thalachh	Maximum heart rate achieved
Slp	Slope
Caa	Number of major vessels
Thall	Thallium stress test result
Target	Likelihood of heart attack: 0 = less chance, 1 = more chance

B. Feature extraction

The feature extraction stage, deemed the most critical stage in machine learning, follows the preprocessing stage. Prior studies have mainly concentrated on devising optimal methods for feature extraction. Common features extracted for diagnosing shockable ECG rhythms are categorized into four groups: temporal/morphological features, spectral features, time-frequency/wavelet features, and complexity features (nonlinear features). Due to the time-varying and unpredictable nature of ECG signals, Variance Inflation Factor (VIF) have gained more attention as an effective feature extraction method for shockable diagnosis in computer-aided diagnosis systems. The VIF quantifies the extent to which the variance of an estimated regression coefficient is increased due to multicollinearity. Specifically, the VIF for a predictor variable is calculated as the ratio of the variance of the coefficient estimate when that variable is included in a regression model to the variance of the coefficient estimate when that variable is not included in the model and the results are shown in Table II.

TABLE II
FEATURE EXTRACTION USING VARIANCE INFLATION FACTOR (VIF)

Feature	VIF
thall	10.628714
slp	7.524873
sex	3.757474
output	3.713174
oldpeak	2.557867
cp	2.512613
restecg	2.135752
exng	1.946123
caa	1.828787
fbs	1.220503

C. Feature selection

The feature selection stage is critical in removing redundant features, reducing computational costs, and enhancing the system's overall performance. This stage employed Pearson Correlation as feature selection method and it is illustrated in Fig. 7.

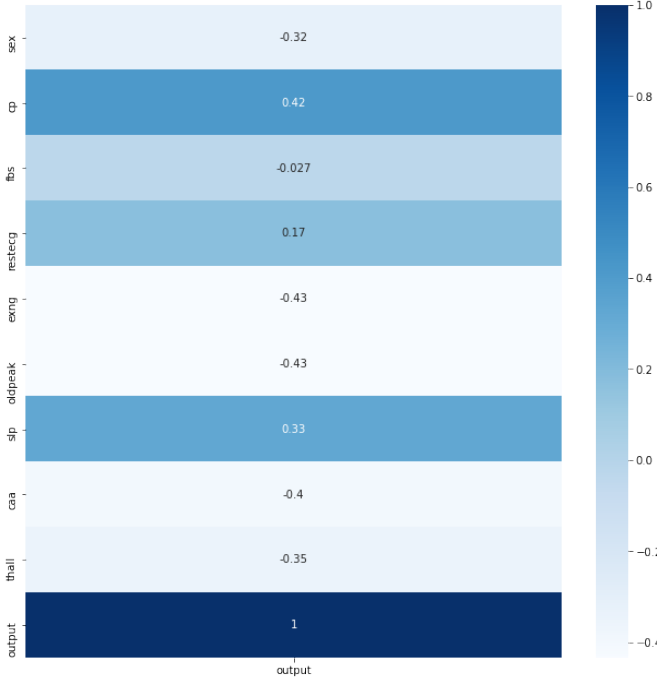


Fig. 7. Feature selection report

D. Features reduction

At this stage, a smaller set of new variables is created, each composed of a blend of input variables that contain the same information as the entire set of input variables. We used Interquartile Range (IQR) method to improve model performance. It is a measure of statistical dispersion that quantifies the spread of a dataset by focusing on the middle 50% of the data. It is defined as the difference between the third quartile (Q3) and the first quartile (Q1).

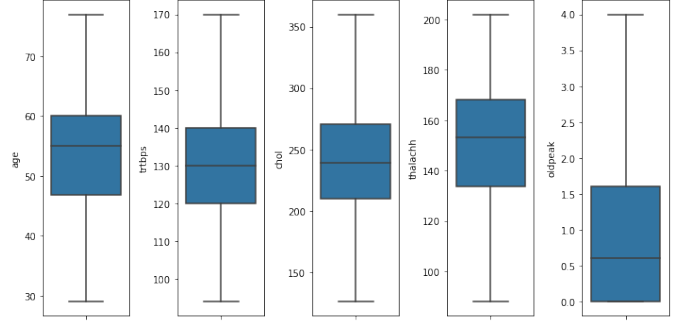


Fig. 8. Interquartile Range result

E. Classification

The final step in ECG signal analysis is classification, where the input ECG signal is categorized after selecting the crucial features. The classification report as shown in Table 9 provides a detailed evaluation of the classification model's performance, indicating its ability to distinguish between two classes (0 and 1). The precision metric, which measures the proportion of true positive predictions out of all positive predictions made by the model, is 0.91 for class 0 and 0.80 for class 1. The recall metric, representing the proportion of true positive instances correctly identified by the model, is 0.74 for class 0 and 0.93 for class 1. The F1-score, a balanced measure of precision and recall, is 0.82 for class 0 and 0.86 for class 1. The support metric indicates the number of instances in each class, with 27 instances for class 0 and 30 instances for class 1. Overall accuracy, reflecting the percentage of correctly classified instances, is 0.84. This comprehensive assessment showcases the model's effectiveness in accurately classifying instances, providing valuable insights into its performance across different evaluation metrics.

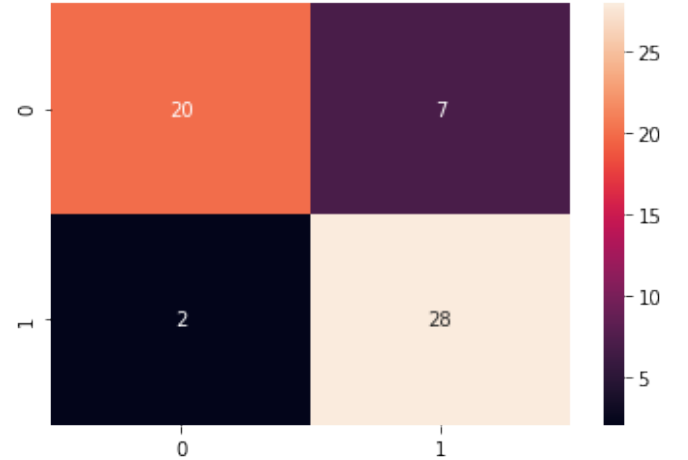


Fig. 9. Classification report

V. SYSTEM LEVEL DESIGN VALIDATION

The human ECG signal comprises PQRST waves, with the frequency range of P and T waves at 0.5 Hz and 10 Hz,

TABLE III
CLASSIFICATION REPORT

Class	Precision	Recall	F1-Score	Support
0	0.91	0.74	0.82	27
1	0.80	0.93	0.86	30
Accuracy	0.84			
Macro avg	0.85	0.84	0.84	57
Weighted avg	0.85	0.84	0.84	57

respectively, and the QRS complex ranging between 4 and 20 Hz. The standard frequency range for ECG signals spans from 0.05 to 100 Hz. While a bandpass filter only permits signals within specific frequencies, a low-pass filter allows signals below the cutoff frequency while attenuating higher frequencies. In this context, a low-pass filter is utilized, as depicted in Figure 1, to process the ECG signal input. It effectively attenuates noise and generates the desired response. An AC analysis of the filter is conducted with a cutoff frequency of 100 Hz.

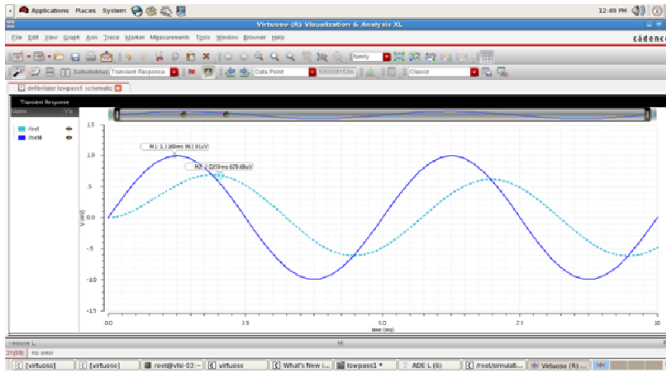


Fig. 10. Low pass filter response (Attenuation of high frequency signal)

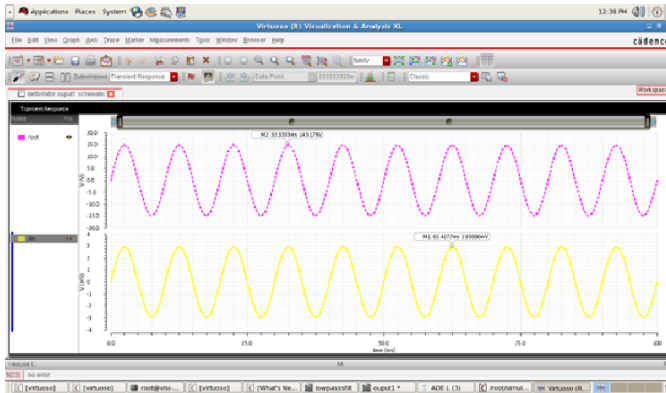


Fig. 11. Simulation of Automatic gain control amplifier

Fig. 12 illustrates the plot of the comparator block. The input signal is set to 7.2V, representing the ECG signal of the monitored person, while the reference signal is set to 3V, representing the standard ECG signal of a normal human being. The input signal is an AC sine wave, while the reference signal is DC. The comparator circuit compares these signals

and generates a binary output. During the positive half cycle, if the input signal surpasses the reference signal, the output is binary low. Conversely, during the negative half cycle, if the input falls below the reference, the output is binary high, with an amplitude of 6.6V. When the comparator output is low, a shock is triggered to help restore the heart's natural rhythm and address cardiac arrest.

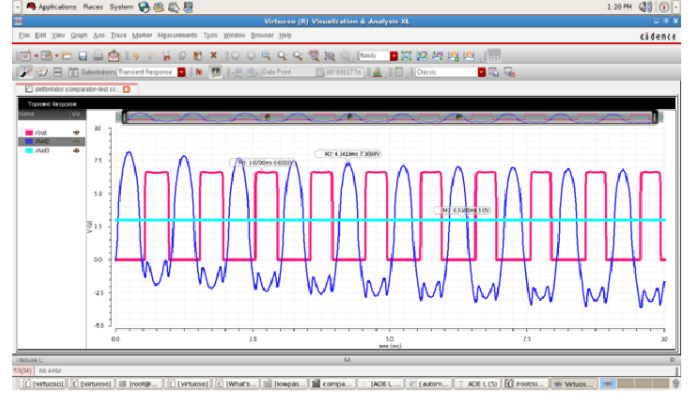


Fig. 12. Comparator Output

VI. CONCLUSION

The project design followed a systematic approach, simulating each step using a modular design method. The architecture was segmented into sub-modules, allowing for individual design and testing to ensure desired outcomes. Subsequently, the blocks were integrated to verify overall functionality. The project aims to automatically administer shocks when the heartbeat rate deviates from the individual's threshold value, eliminating the need for manual intervention or hospital assistance and potentially reducing victim mortality rates. A low-pass filter is employed to suppress frequencies above 1.67 Hz, corresponding to the standard heartbeat frequency, followed by automatic gain control amplification to boost signal levels. The amplified signal is then processed by a Comparator to generate the desired output. The project also includes post-project enhancements and documentation of block outputs. The therapeutic unit, the defibrillating unit, incorporates a PWM circuit block, which will be designed and simulated in subsequent project stages. The DL model is used for the classification of the signal. Future work will involve interfacing with real-time ECG signals instead of sinusoidal signals, given their similarity to ECG waveforms.

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